Solar Panels and Photovoltaic Energy

Knowledge document 013 of 100

Introduction to Solar Energy

Solar energy is energy from the Sun, which can be converted into electricity using photovoltaic (PV) technology or into heat using solar thermal systems. The Sun delivers far more energy to the Earth in an hour than humanity uses in a year, and sunlight is free, clean and renewable. A solar panel, also called a PV module, is made up of many connected solar cells. Solar power does not emit carbon dioxide while generating electricity, which makes it central to fighting climate change. India, with about 300 sunny days in most regions, has excellent solar potential.

History

The photovoltaic effect was discovered by the French physicist Edmond Becquerel in 1839. In 1954 scientists Daryl Chapin, Calvin Fuller and Gerald Pearson at Bell Laboratories created the first practical silicon solar cell, with about 6 percent efficiency. Solar cells powered the Vanguard 1 satellite launched in 1958. Since then, panel prices have fallen by more than 90 percent, making solar the cheapest source of new electricity in many countries.

How Does a Solar Cell Produce Electricity?

A solar cell is made of semiconductor material, usually silicon, with a p-n junction formed by joining n-type and p-type layers. When sunlight, made of tiny energy packets called photons, strikes the cell, photons with enough energy knock electrons loose from silicon atoms, creating free electrons and holes. This is the photovoltaic effect. The electric field at the p-n junction pushes electrons toward the n-side and holes toward the p-side, so a voltage is created. When metal contacts on the front and back of the cell are connected to a circuit, the electrons flow through it as direct current (DC). A single silicon cell produces only about 0.5 to 0.6 volts, so around 60 to 72 cells are wired in series to make a panel. Because homes use alternating current (AC), an inverter converts the DC output into AC. A charge controller and battery can be added for storage.

Types of Solar Panels

Monocrystalline panels are made from single-crystal silicon, are black in colour, and are the most efficient, commonly 19 to 23 percent. Polycrystalline panels use many silicon crystals, are blue, and have efficiencies of about 15 to 18 percent, though they have become less common. Thin-film panels use materials such as cadmium telluride, are lightweight and flexible but less efficient. Newer technologies include perovskite cells and tandem cells that stack perovskite on silicon, which have reached laboratory efficiencies above 30 percent.

What Is Solar Panel Efficiency?

Solar panel efficiency is the percentage of sunlight energy falling on the panel that is converted into usable electricity. If a panel receives 1,000 watts of sunlight per square metre and converts 200 watts into electricity, its efficiency is 20 percent. Higher efficiency means more power from the same roof area. Efficiency is affected by temperature (panels perform slightly worse when very hot), shading, dust, the angle of the panel, and panel quality. Panels are rated in watts peak (Wp) under standard test conditions, and a typical modern panel is rated about 400 to 600 Wp. Panels degrade slowly, about 0.5 percent per year, and typically last 25 years or more. In India panels should face south and be tilted roughly at an angle close to the local latitude.

Types of Solar Power Systems

A grid-connected (on-grid) system feeds electricity to the utility grid and has no battery. An off-grid system works independently with batteries and is used in remote areas. A hybrid system combines grid connection and batteries.

What Is Net Metering?

Net metering is a billing arrangement that allows a consumer with a rooftop solar system to send surplus electricity to the grid and receive credit for it. A bidirectional meter records both the electricity imported from the grid and the electricity exported. At the end of the billing period, the consumer pays only for the net consumption, which is units imported minus units exported. If more is exported than imported, the surplus units are carried forward as credit. Net metering reduces electricity bills and encourages households and businesses to install solar panels. Tamil Nadu and other Indian states have net metering rules for rooftop solar systems.

Solar Energy in India

India launched the Jawaharlal Nehru National Solar Mission in 2010. The country crossed 100 gigawatts of installed solar capacity in early 2025 and aims for 500 GW of non-fossil capacity by 2030. The Bhadla Solar Park in Rajasthan is among the world's largest solar parks, with about 2,245 MW capacity. Tamil Nadu's Kamuthi Solar Power Project, commissioned in 2016, has a capacity of 648 MW. Cochin International Airport became the world's first fully solar-powered airport in 2015. In February 2024 the government launched the PM Surya Ghar: Muft Bijli Yojana to install rooftop solar on one crore homes, with subsidies for households. India and France founded the International Solar Alliance in 2015.

Challenges and Future

Challenges include dependence on sunlight, the need for storage, land requirements, panel recycling and imported cells. The future includes cheaper batteries, floating solar farms, agrivoltaics and perovskite panels.

Quick Facts

- A solar cell converts sunlight to DC electricity through the photovoltaic effect at a p-n junction.
- Efficiency is the percentage of sunlight converted to electricity; good panels are 19 to 23 percent efficient.
- Net metering credits consumers for surplus solar power sent to the grid.
- The photovoltaic effect was discovered by Edmond Becquerel in 1839.
- India's PM Surya Ghar scheme was launched in February 2024 for rooftop solar on one crore homes.